

$$\begin{aligned}
& \max && x_1 &+& 3x_2 \\
& \text{subject to} && x_1 &+& 2x_2 \leq 3 \\
& && x_1 && \geq 1 \\
& && x_1, x_2 \geq 0
\end{aligned}$$

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad c = (1 \quad 3)$$

$$A = \begin{pmatrix} -1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

$$\begin{aligned}
& \text{Find a vector} && \mathbf{x} \\
& \text{that maximizes} && \mathbf{c}^\top \mathbf{x} \\
& \text{subject to} && A\mathbf{x} \leq \mathbf{b} \\
& \text{and} && \mathbf{x} \geq \mathbf{0}.
\end{aligned}$$

$$f_j = 1(0) \text{ if we do(don't) travel on road } j$$

$$w_j = \text{cost (time) to travel along road } j$$

For every node i

$$\sum_{j \in E} A_{ij} \cdot f_j = \begin{cases} -1 & \text{if } i \text{ is the starting node} \\ 1 & \text{if } i \text{ is the final node} \\ 0 & \text{otherwise} \end{cases}$$

$$\min \sum_{j \in E} w_j \cdot f_j$$

subject to

$$\sum_{j \in E} A_{ij} \cdot f_j = \begin{cases} -1 & \text{if } i \text{ is the starting node} \\ 1 & \text{if } i \text{ is the final node} \\ 0 & \text{otherwise} \end{cases}$$

$$f_j = \begin{cases} 1 & \text{if we travel along road } j \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{r \in R} f_r \leq B$$

R is the set of all "red-nodes", B is the maximum of these nodes we can travel through.

$$\sum_{j: (i,j) \in E} f_{ij} - \sum_{j: (j,i) \in E} f_{ji} = \begin{cases} 1 & \text{if } i = s \\ -1 & \text{if } i = t \\ 0 & \text{otherwise} \end{cases}$$

$$f_{ij} \in \{0, 1\} \quad \forall (i, j) \in E.$$